

# Contextuality and Quantum Advantage in Small Non-local Games

## Progress and Research Directions

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September 2026 update.

## Overview

As of September 2026, the project includes a computational pipeline for exploring contextuality and quantum advantage in small non-local games. This note sets out what it does, what the results suggest, and a possible direction for discussion. It also brings the main ideas together in one place; the companion CHSH walkthrough contains detailed worked calculations of this game through the pipeline.<sup>1</sup>

The project uses the sheaf-theoretic account of contextuality and the contextual fraction as its quantitative measure [1, 2]. It combines systematic enumeration of games with searches for successful strategies, then asks what explains the patterns found.

The current work asks three questions:

1. Which games admit an advantage over classical strategies?
2. How much contextuality is required to attain a given score?
3. Under noise, when does a strategy cease to be contextual, and when does it cease to outperform classical strategies in the game being played?

**Current position.** The saved two-party catalogue contains 805 classes of binary games, of which nine have a no-signalling advantage. Quantum searches find an advantage for all nine. The contextual-fraction decomposition bounds the failure probability by

$$p_{\text{fail}}(e) \geq \text{NCF}(e)(1 - \omega_c) + \text{CF}(e)(1 - \omega_{\text{ns}}).$$

Some located strategies attain this bound numerically; six two-party classes retain contextuality above the minimum required by their achieved scores in the searches performed so far.

One possible direction is to understand this equality question: *when can quantum strategies achieve their best game score with only the contextuality required by the bound?* The current data identify concrete examples on which to investigate it. Noise calculations provide a second perspective: contextuality can remain after an advantage in a particular game has disappeared.

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<sup>1</sup>The CHSH Walkthrough should be finished by mid October.

# 1 Framework and scope

## 1.1 Games and behaviours

In a non-local game, separated players receive questions and return answers without communicating. The games studied here give each player two possible questions and two possible answers, labelled 0 and 1. A *context* is a combination of questions: for two players, these are 00, 01, 10 and 11.

The implementation represents a game by its winning table and a strategy's behaviour by its probability table. Both use questions as rows and answers as columns:

| Table                            | Entries                   | Interpretation                                                    |
|----------------------------------|---------------------------|-------------------------------------------------------------------|
| Rules<br>$V(ab \mid xy)$         | 1 for a win; 0 for a loss | Whether answers $ab$ win when the questions are $xy$ .            |
| Probabilities<br>$e(ab \mid xy)$ | A number between 0 and 1  | How likely the players are to give answers $ab$ when asked $xy$ . |

Each row of  $e$  is a distribution over answers  $ab$  conditional on questions  $xy$ , and sums to one. This table is the *behaviour*, or *empirical model*, associated with a strategy. Using probability tables makes it possible to compare classical, quantum and no-signalling behaviours within the same framework, independently of their physical implementation.

The score is the probability assigned to winning answers, averaged over the questions. All question combinations in the full-context catalogues are equally likely. For  $n$  players this gives

$$p_{\text{win}}(e) = 2^{-n} \sum_{x,a} V(a \mid x) e(a \mid x), \quad p_{\text{fail}}(e) = 1 - p_{\text{win}}(e). \quad (1)$$

Here  $x$  and  $a$  stand for the complete lists of questions and answers. For example, if a row of the rules accepts answers 00 and 11, the chance of winning on that row is  $e(00 \mid xy) + e(11 \mid xy)$ .

## 1.2 Classical, quantum and no-signalling values

The *value* of a game is its best possible winning probability under a specified set of allowed behaviours:

| Value                               | Allowed behaviours                                                                                                     |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Classical, $\omega_c$               | Local response plans, possibly chosen using shared randomness agreed before the questions arrive.                      |
| Quantum, $\omega_q$                 | Local measurements on a shared quantum state.                                                                          |
| No-signalling, $\omega_{\text{ns}}$ | Probability tables in which the statistics seen by any group of players do not depend on the other players' questions. |

These sets are nested, so  $\omega_c \leq \omega_q \leq \omega_{\text{ns}}$ . No-signalling permits some tables that quantum mechanics cannot produce. Its value is therefore a useful upper limit on quantum performance.

**Quantities compared in the project.** The rules and question distribution determine  $\omega_c$ ,  $\omega_q$  and  $\omega_{\text{ns}}$ . Each behaviour  $e$  has its own score and contextual fraction. The analysis compares these properties of the located behaviours with the limits imposed by the game.

### 1.3 Enumerated game families

Alongside the general two-party catalogue, the project examines XOR games, whose winning rule specifies the parity of the players' answers. The full-context XOR family imposes a parity requirement on every possible question combination. This gives a small, structured family that can be enumerated completely for two and three parties.

For two players,  $a \oplus b = 0$  means the answers agree;  $a \oplus b = 1$  means they differ. CHSH is an example. Its **rules table** is:

| Questions $xy$ | Answers $ab$ |    |    |    | Required parity |
|----------------|--------------|----|----|----|-----------------|
|                | 00           | 01 | 10 | 11 |                 |
| 00             | 1            | 0  | 0  | 1  | 0               |
| 01             | 1            | 0  | 0  | 1  | 0               |
| 10             | 1            | 0  | 0  | 1  | 0               |
| 11             | 0            | 1  | 1  | 0  | 1               |

Each row has one of two possible patterns. Choosing a pattern independently for each of four rows gives  $2^4 = 16$  labelled two-party XOR games.

For three players,  $a \oplus b \oplus c = 0$  accepts answers with an even number of 1s: 000, 011, 101 and 110. Parity 1 accepts the other four answers. There are eight question combinations and eight answer combinations. For example, requiring even parity everywhere except on question 111 gives:

| Questions $xyz$ | Answers $abc$ |     |     |     |     |     |     |     |
|-----------------|---------------|-----|-----|-----|-----|-----|-----|-----|
|                 | 000           | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
| 000             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 001             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 010             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 011             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 100             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 101             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 110             | 1             | 0   | 0   | 1   | 0   | 1   | 1   | 0   |
| 111             | 0             | 1   | 1   | 0   | 1   | 0   | 0   | 1   |

There are  $2^8 = 256$  such labelled games. As in the CHSH table, each 1 marks a winning answer combination.

The completed enumerations are summarised below. Games that differ only by renaming players, local questions or local answers are grouped into a class.

| Family                              | Labelled games    | Classes |
|-------------------------------------|-------------------|---------|
| All two-party binary winning tables | $2^{16} = 65,536$ | 805     |
| Full-context two-party XOR games    | 16                | 2       |
| Full-context three-party XOR games  | 256               | 5       |

General games can have any pattern of winning cells. The GHZ–Mermin benchmark constrains four of the eight three-party contexts. Filling its other rows with automatic wins gives a useful separate example, outside the full-context XOR family. General three-party games have not been exhaustively enumerated.

## 1.4 Contextual fraction and local decomposition

In these Bell scenarios, a non-contextual behaviour is a local behaviour: its probability table can be explained using local response plans and shared randomness [1]. The contextual fraction asks how much of a given table cannot be accounted for in that way.

The measure is defined through a decomposition of the behaviour into a local contribution and a no-signalling remainder:

$$e = \alpha e_L + (1 - \alpha) e_R. \quad (2)$$

Both components are complete, normalised probability tables on the same scenario as  $e$ :

- $e_L$  is a **local behaviour**: the players can produce the whole table by agreeing on a randomly selected local response plan.
- $e_R$  is the **remaining behaviour**: another valid no-signalling table that completes the mixture. It need not be achievable in quantum mechanics, even if  $e$  is.

The mixture is taken cell by cell, with the same weight  $\alpha$  throughout. For example, at  $\alpha = 0.6$ , entries of 0.3 and 0.5 in the component tables give  $0.6(0.3) + 0.4(0.5) = 0.38$  in the combined table. A valid decomposition must reproduce the entire behaviour consistently in this way.

In the contextual-fraction framework [2], we take the *largest possible* local weight  $\alpha$ . This is the non-contextual fraction; its complement is the contextual fraction:

$$e = \text{NCF}(e) e_L + \text{CF}(e) e_R, \quad \text{CF}(e) = 1 - \text{NCF}(e). \quad (3)$$

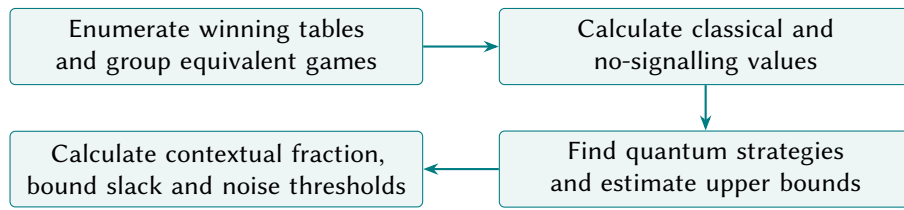
At this maximum, a positive-weight remainder is strongly contextual: there is no further local fraction to extract from it. The two component tables can be non-unique, even though the maximum weight is fixed.

**Connection to game performance.** Each component has its own winning probability. The decomposition therefore expresses the score of  $e$  as a weighted average of a local score and a no-signalling score. This is the starting point for the performance bound in Section 4.

At  $\text{CF}(e) = 0$ , the whole table is local. At  $\text{CF}(e) = 1$ , it contains no local fraction. Intermediate values quantify the largest local contribution compatible with the full probability table. This is a mathematical decomposition of the behaviour; it need not describe how an experiment physically produces its outcomes.

The project uses this measure to relate the contextuality of a strategy to its performance, and to distinguish contextuality that is necessary for a given score from any additional contextuality in the behaviour found.

## 2 Computational approach and checks



### 2.1 Classical values

The code checks every deterministic local response plan. For two parties, one plan specifies Alice’s answer to each of her two possible questions and Bob’s answer to each of his. Evaluating these plans gives the best classical winning probability.

Shared randomness allows mixtures of these plans, but their scores combine as weighted averages. Such an average cannot exceed the best individual score, so checking deterministic plans is sufficient. With  $m$  equally likely contexts and at most  $k$  winnable by a single plan,  $\omega_c = k/m$ .

### 2.2 Linear programming

Linear programming (LP) is used for two different tasks: finding the best no-signalling score of a game, and measuring the contextuality of a fixed behaviour. Both have linear objectives and constraints, but their unknowns are different. CHSH provides a concrete example of each.

**The incidence matrix and local weights.** For the contextual-fraction calculation, the probability table  $e$  is already given. Flattening its four rows produces a vector  $\mathbf{v}$  of 16 probabilities. Each row sums to one, so the entries of  $\mathbf{v}$  sum to four, not one.

A deterministic local plan  $g = (A_0, A_1, B_0, B_1)$  fixes an answer to every possible local question. There are  $2^4 = 16$  such plans. The *incidence matrix*  $M$  records their predictions: its rows correspond to table cells and its columns to plans, with an entry of one when the plan selects that cell and zero otherwise. For CHSH,  $M$  is  $16 \times 16$ ; each column has four ones, one selected cell per context.

The vector  $\mathbf{b}$  assigns a nonnegative weight to each plan. Multiplying by  $M$  adds their contributions to each cell. A completely local table can be written as  $M\mathbf{b} = \mathbf{v}$  with total weight one. For a contextual table, the LP instead fits the largest *partial* local contribution beneath it:

$$\text{NCF}(e) = \max_{\mathbf{b} \geq 0} \mathbf{1}^T \mathbf{b} \quad \text{subject to} \quad M\mathbf{b} \leq \mathbf{v}, \quad \text{CF}(e) = 1 - \text{NCF}(e).$$

Here  $\mathbf{1}$  is the all-ones vector and the inequality is cell by cell. The weights are allowed to sum to less than one; requiring a total of one would leave no feasible solution for a contextual table.

At an optimum of total weight  $\alpha$ ,  $M\mathbf{b}$  contributes  $\alpha$  to each context’s row sum. If  $\alpha > 0$ , dividing it by  $\alpha$  gives the normalised local component  $e_L$ . If  $\alpha < 1$ , the remaining vector  $(\mathbf{v} - M\mathbf{b})/(1 - \alpha)$  gives  $e_R$ . The code implements this in `noncontextual_fraction`; `contextual_fraction` returns its complement.

**Optimising over no-signalling tables.** To find  $\omega_{\text{ns}}$ , the game rules are fixed and the whole probability table is unknown. Write its entries as  $p(ab \mid xy)$  and collect them in a vector  $\mathbf{p}$ . With CHSH's four equally likely contexts, the LP is

$$\omega_{\text{ns}} = \max_{\mathbf{p}} \mathbf{w}^\top \mathbf{p}, \quad w_{xyab} = \frac{1}{4} V(ab \mid xy).$$

Thus each winning cell has objective coefficient  $1/4$  and each losing cell has coefficient zero. The constraints describe valid no-signalling probabilities:

$$\begin{aligned} p(ab \mid xy) &\geq 0, & \sum_{a,b} p(ab \mid xy) &= 1 & \text{for every } x, y, \\ \sum_b p(ab \mid x0) &= \sum_b p(ab \mid x1) & & & \text{for every } x, a, \\ \sum_a p(ab \mid 0y) &= \sum_a p(ab \mid 1y) & & & \text{for every } y, b. \end{aligned}$$

The last two lines say that Alice's marginal distribution is independent of Bob's question, and vice versa. For example,

$$p(00 \mid 00) + p(01 \mid 00) = p(00 \mid 01) + p(01 \mid 01)$$

keeps Alice's probability of answering zero to question zero unchanged when Bob changes his question.

The code collects these normalisation and marginal equations into  $A_{\text{eq}} \mathbf{p} = \mathbf{b}_{\text{eq}}$ . This constraint matrix has a different role from the incidence matrix  $M$ : it enforces consistency between probabilities rather than expressing a mixture of local plans. No local decomposition is required.

For CHSH, the following table is feasible:

| Questions $xy$ | $ab = 00$ | 01    | 10    | 11    |
|----------------|-----------|-------|-------|-------|
| 00             | $1/2$     | 0     | 0     | $1/2$ |
| 01             | $1/2$     | 0     | 0     | $1/2$ |
| 10             | $1/2$     | 0     | 0     | $1/2$ |
| 11             | 0         | $1/2$ | $1/2$ | 0     |

Each player's marginal is  $1/2$  for either answer, while every nonzero cell wins. This is the PR box. It attains a score of one, proving  $\omega_{\text{ns}} = 1$  for CHSH, since no winning probability can exceed one. It is no-signalling but is not quantum-realizable. For other games, `no_signalling_value` carries out the same optimisation with their winning-cell coefficients.

**Two distinct outputs.** The no-signalling LP varies the probability table to find a game's best possible no-signalling score. The contextual-fraction LP keeps a table fixed and varies the weights of local plans to measure its contextuality. The dual below belongs to this second calculation.

## 2.3 Bell inequalities and dual witnesses

The contextual-fraction LP above is called the *primal*. It constructs a local contribution that fits inside the given table. Its dual approaches the same problem from the other direction: it provides an upper bound on how much local weight can fit. Agreement between the two certifies the optimum; the implementation solves them numerically.

**Weights that test every local plan.** The dual assigns a nonnegative test weight  $z_i$  to each probability-table cell. These weights are coefficients, not probabilities: they need not sum to one. It solves

$$\min_{z \geq 0} z^\top v \quad \text{subject to} \quad M^\top z \geq \mathbf{1}.$$

The transpose  $M^\top$  reverses how the incidence matrix is used. For each deterministic plan, it adds the test weights of the cells that plan selects. The constraint requires that sum to be at least one for *every* plan.

This gives a bound on any feasible primal weights  $b$ :

$$\underbrace{\mathbf{1}^\top b}_{\text{local weight}} \leq z^\top M b \leq \underbrace{z^\top v}_{\text{dual bound}}.$$

The first inequality follows because each plan has test weight at least one. The second follows because its combined contribution stays below  $v$  cell by cell and the test weights are nonnegative. The dual searches for the smallest such upper bound. LP duality ensures that it meets the primal maximum:

$$\min_{\substack{z \geq 0 \\ M^\top z \geq \mathbf{1}}} z^\top v = \text{NCF}(e).$$

**From a bound to a Bell inequality.** A normalised local behaviour is a mixture of deterministic plans with total weight one. Since every plan has test weight at least one, every such mixture obeys

$$z^\top p \geq 1 \quad \text{for every local probability table } p.$$

This is a Bell inequality: a linear restriction respected by all local behaviours. If the observed table gives  $z^\top v < 1$ , its violation rules out a local explanation. The coefficients  $z$  are therefore a *witness* of contextuality. A feasible dual solution already supplies a witness when it gives a value below one; an optimal solution also recovers  $\text{CF}(e) = 1 - z^\top v$ .

The implementation in `bell_inequality` rewrites this with an upper bound of zero. For  $m$  contexts it assigns each cell the coefficient  $c_i = 1/m - z_i$ , or, in vector form,  $c = \mathbf{1}/m - z$ . Each context's probability row sums to one. Since  $p$  contains all  $m$  rows, its entries sum to  $m$ :  $\sum_i p_i = m$ . Therefore, for any normalised probability table,

$$\begin{aligned} c^\top p &= \sum_i c_i p_i = \sum_i \left( \frac{1}{m} - z_i \right) p_i \\ &= \frac{1}{m} \sum_i p_i - \sum_i z_i p_i \\ &= \frac{1}{m} m - z^\top p \\ &= 1 - z^\top p. \end{aligned}$$

For a local table,  $z^\top p \geq 1$ , so

$$c^\top p = 1 - z^\top p \leq 0.$$

For a contextual behaviour, the optimal witness has normalised violation equal to  $\text{CF}(e)$  [2].

**CHSH as an explicit witness.** For CHSH, choose test weight one on every losing cell and zero on every winning cell. In table form the coefficients are

| Questions $xy$ | $ab = 00$ | 01 | 10 | 11 |
|----------------|-----------|----|----|----|
| 00             | 0         | 1  | 1  | 0  |
| 01             | 0         | 1  | 1  | 0  |
| 10             | 0         | 1  | 1  | 0  |
| 11             | 1         | 0  | 0  | 1  |

Every deterministic local plan loses at least one of the four contexts, so  $M^T \mathbf{z} \geq \mathbf{1}$ : these coefficients are dual-feasible. Their value on a behaviour adds its four conditional losing probabilities. Since contexts are equally likely,

$$\mathbf{z}^T \mathbf{v} = 4p_{\text{fail}}(e).$$

The resulting Bell inequality is therefore the usual CHSH winning bound:

$$4p_{\text{fail}}(e) \geq 1 \iff p_{\text{win}}(e) \leq \frac{3}{4} \quad \text{for every local behaviour.}$$

A score above  $3/4$  violates this inequality. Moreover, the dual bound on local weight implies

$$\text{NCF}(e) \leq 4p_{\text{fail}}(e), \quad \text{CF}(e) \geq 1 - 4p_{\text{fail}}(e) = 4p_{\text{win}}(e) - 3.$$

For the standard optimal quantum CHSH table,  $p_{\text{win}} = (2 + \sqrt{2})/4 \simeq 0.853553$ . This witness gives

$$\mathbf{z}^T \mathbf{v} = 2 - \sqrt{2} \simeq 0.585786.$$

The primal reaches that same local weight: assigning weight  $(2 - \sqrt{2})/8$  to each of the eight plans that win three contexts produces a local contribution that fits beneath this table. The matching values show that both solutions are optimal, giving

$$\text{NCF}(e) = 2 - \sqrt{2}, \quad \text{CF}(e) = \sqrt{2} - 1 \simeq 0.414214.$$

In the code's upper-bound convention, the same inequality reads  $4p_{\text{win}} - 3 \leq 0$ . Its maximum possible value over arbitrary normalised probability tables is one, so its violation at this quantum table,  $\sqrt{2} - 1$ , is already normalised.

CHSH is a case where the game's own score supplies an optimal witness. For a general game and behaviour, the dual searches over all admissible cell weights, so its best witness may differ from that game's winning-score inequality. Comparing them can reveal contextuality that the particular game score does not fully capture.

## 2.4 Quantum strategies and upper bounds

Quantum performance is assessed from two directions. First, a search over measurement settings for one qubit per party constructs explicit strategies. For fixed measurements, the game operator identifies the best shared state. Repeating the search from several starting settings gives achievable scores and hence numerical lower estimates of the quantum value.

Second, a semidefinite programme estimates an upper bound using the NPA approach [3]. It optimises over a matrix of expectation values, subject to consistency and positivity conditions obeyed by quantum measurements. The matrix is indexed by selected products of measurement operators, often called *words*. The current two-party calculation uses

$$\{I, A_0, A_1, B_0, B_1, A_0B_0, A_0B_1, A_1B_0, A_1B_1\}.$$

Here  $A_x$  and  $B_y$  are the players' observables and  $I$  is the identity. The matrix entry indexed by  $u, v$  represents  $\langle u^\dagger v \rangle$ . Using at most one observable per party gives a  $9 \times 9$  matrix for two parties and a  $27 \times 27$  matrix for three.

These conditions can admit more than the quantum set, which is why the optimisation gives an upper bound. Comparing it with explicit-strategy scores indicates how closely the searches approach the optimum, subject to the numerical limitations below.

## 2.5 Validation and numerical limitations

All 46 regression tests passed on 17 September 2026, including CHSH, GHZ–Mermin, contextual-fraction witnesses and noise benchmarks.

The file `catalogue_quantum.csv` contains 16 saved results, each for a representative game. Each record gives the classical and no-signalling values, the quantum score found, and the numerical quantum upper estimate. For all 16 records, the lower and upper estimates agree within  $1.4 \times 10^{-8}$ . All saved upper estimates are slightly below the corresponding lower estimates, which signals numerical error. The code clips negative gaps to zero before setting its `certified` flag. **That flag is not a rigorous mathematical certificate.** The results are therefore reported as achieved scores  $q_{\text{found}}$ , with independent verification of optimality still needed. Input checks and solver-error handling also need review. There are 15 distinct classes in these tables: CHSH is listed in both the general and XOR catalogues.

### 3 Game values and classification

#### Catalogue results

The 65,536 two-party winning tables reduce to 805 classes after grouping games that differ only by the allowed relabellings. The saved class sizes account for all 65,536 tables. Nine classes have a no-signalling value above their classical value, and quantum searches find an advantage for all nine.

| Class ID     | $\omega_c$ | $q_{\text{found}}$ | $\omega_{\text{ns}}$ | $\text{CF}(e_{\text{found}})$ |
|--------------|------------|--------------------|----------------------|-------------------------------|
| 5158         | 1/2        | 0.544599           | 5/8                  | 0.379787                      |
| 5166         | 1/2        | 0.551777           | 5/8                  | 0.414214                      |
| 5734         | 3/4        | 0.782218           | 7/8                  | 0.310839                      |
| 5735         | 3/4        | 0.786559           | 7/8                  | 0.340981                      |
| 5817         | 3/4        | 0.788675           | 7/8                  | 0.347151                      |
| 5821         | 3/4        | 0.794599           | 7/8                  | 0.379787                      |
| 7606         | 3/4        | 0.794599           | 7/8                  | 0.379787                      |
| 7614         | 3/4        | 0.801777           | 7/8                  | 0.414214                      |
| 26217 (CHSH) | 3/4        | 0.853553           | 1                    | 0.414214                      |

Table 1. The nine two-party classes with a recorded no-signalling advantage. IDs label representatives under the code’s encoding convention.

Class 5158 illustrates the comparison: classical strategies can win half the time. The quantum behaviour found wins about 54.46% of the time, below the no-signalling value of 62.5%. That particular behaviour has contextual fraction about 0.380. The CF column describes the located probability table  $e_{\text{found}}$ , not all strategies for that game.

CHSH has the largest classical-to-no-signalling gap in this catalogue, 1/4. Among the nine advantaged classes it alone has  $\omega_{\text{ns}} = 1$ . Classically satisfiable games elsewhere in the catalogue also have  $\omega_{\text{ns}} = 1$ , so that observation is specific to the advantaged group.

#### 3.1 The three-party XOR catalogue

The five classes are labelled X1–X5 below, with full identifiers in Appendix A.

| Class | $\omega_c$ | $q_{\text{found}}$ | $\omega_{\text{ns}}$ | Observed pattern              |
|-------|------------|--------------------|----------------------|-------------------------------|
| X1    | 1          | 1.000000           | 1                    | Classically satisfiable       |
| X2    | 7/8        | 0.875000           | 1                    | No detected quantum advantage |
| X3    | 3/4        | 0.853553           | 1                    | Quantum advantage             |
| X4    | 3/4        | 0.750000           | 1                    | No detected quantum advantage |
| X5    | 3/4        | 0.853553           | 1                    | Quantum advantage             |

For X2 and X4, the searches and upper estimates agree numerically with the classical value, despite the higher no-signalling value. For the four non-trivial classes, they also indicate that perfect quantum success is unavailable. These are observations supported by closely matching numerical estimates, not exact proofs for this catalogue or for XOR games in general.

The GHZ–Mermin benchmark behaves differently: it has a perfect quantum strategy on its four constrained contexts and classical value 3/4. Adding four automatic-win contexts preserves perfect quantum success and raises the classical value to 7/8. This gives a concrete reason to be interested in which contexts carry constraints.

## 4 Connecting contextuality to game performance

### 4.1 Resource bound and slack

Abramsky, Barbosa and Mansfield relate a strategy's failure probability to its non-contextual fraction [2]. For the uniformly sampled games here, their resource bound takes the form

$$p_{\text{fail}}(e) \geq \text{NCF}(e)(1 - \omega_c). \quad (4)$$

The best local behaviour still fails with probability  $1 - \omega_c$ . The local fraction of  $e$  must contribute at least its share of that failure.

The *slack* is the amount by which actual failure exceeds this bound:

$$\text{original slack} = p_{\text{fail}}(e) - \text{NCF}(e)(1 - \omega_c).$$

For example, a failure probability of 20% against a lower bound of 15% gives slack  $0.20 - 0.15 = 0.05$ . Zero slack means equality; we say the bound is *tight* or *saturated* at that behaviour. Positive slack means the bound does not account for all of its failure. It does not, by itself, mean the strategy can be improved: an optimal quantum strategy can still have positive slack.

### 4.2 Refinement using the no-signalling value

The bound can be refined by retaining information about the no-signalling remainder. Writing  $\alpha = \text{NCF}(e)$  and  $1 - \alpha = \text{CF}(e)$ , the decomposition from Section 1 is

$$e = \alpha e_L + (1 - \alpha) e_R.$$

Because this is a cell-by-cell weighted average, the failure probability is the same weighted average:

$$p_{\text{fail}}(e) = \alpha p_{\text{fail}}(e_L) + (1 - \alpha) p_{\text{fail}}(e_R). \quad (5)$$

Each component has its own minimum failure probability:

$$\begin{array}{ll} p_{\text{fail}}(e_L) \geq 1 - \omega_c & \text{because } e_L \text{ is local,} \\ p_{\text{fail}}(e_R) \geq 1 - \omega_{\text{ns}} & \text{because } e_R \text{ is no-signalling.} \end{array}$$

Both weights are nonnegative, so substituting these limits into (5) gives

$$p_{\text{fail}}(e) \geq \alpha(1 - \omega_c) + (1 - \alpha)(1 - \omega_{\text{ns}}).$$

In terms of the contextual and non-contextual fractions, this is

$$p_{\text{fail}}(e) \geq \text{NCF}(e)(1 - \omega_c) + \text{CF}(e)(1 - \omega_{\text{ns}}). \quad (6)$$

**Contribution of the remainder.** The original bound uses only the fact that the remainder's failure probability is at least zero. The strengthened bound also counts the failure that even a no-signalling behaviour cannot avoid.

### 4.3 Implications for the observed tightness patterns

For a contextual behaviour playing a game with  $\omega_{\text{ns}} < 1$ , the extra term is positive. The original bound therefore has positive slack even before asking whether the quantum strategy is optimal. Every strategy beating  $\omega_c$  is contextual, so this explains the positive original-bound slack for the eight advantaged non-XOR two-party classes.

For XOR games with at least two parties, a no-signalling behaviour can win perfectly. On each row, distribute probability equally over the answers with the required parity: two winning answers for two parties, or four for three. Every proper subset of players then sees uniform answers, independent of the other questions, so the table is no-signalling. Thus  $\omega_{\text{ns}} = 1$  and the extra term vanishes. This does not say that the table is quantum-realizable, or that a quantum strategy attains the bound.

The slack relative to the strengthened bound is

$$\text{strengthened slack} = p_{\text{fail}}(e) - [\text{NCF}(e)(1 - \omega_c) + \text{CF}(e)(1 - \omega_{\text{ns}})].$$

The column named `slack` in the saved CSV uses the *original* bound. Both slacks are mathematically nonnegative; tiny negative numerical values can occur because the calculations use finite precision.

### 4.4 Contextuality required at a given score

Rearranging the strengthened bound gives a lower limit on the contextuality needed to reach a specified winning probability. For game  $G$  and score  $s = p_{\text{win}}(e)$ , the success form of the bound is

$$\begin{aligned} s &\leq \text{NCF}(e)\omega_c + \text{CF}(e)\omega_{\text{ns}} \\ &= (1 - \text{CF}(e))\omega_c + \text{CF}(e)\omega_{\text{ns}} \\ &= \omega_c + \text{CF}(e)(\omega_{\text{ns}} - \omega_c). \end{aligned} \tag{7}$$

For  $\omega_{\text{ns}} > \omega_c$ , this implies

$$\text{CF}(e) \geq \frac{s - \omega_c}{\omega_{\text{ns}} - \omega_c}.$$

Combining this with  $\text{CF}(e) \geq 0$  defines a contextuality floor, denoted  $F_G(s)$ :

$$F_G(s) := \max \left\{ 0, \frac{s - \omega_c}{\omega_{\text{ns}} - \omega_c} \right\}, \quad \text{CF}(e) \geq F_G(p_{\text{win}}(e)). \tag{8}$$

The numerator is the score's improvement over the classical value. The denominator is the full improvement allowed by no-signalling. Their ratio is the fraction of that available improvement needed to reach  $s$ . If  $s \leq \omega_c$ , this formula supplies only the bound zero; it does not say that the particular behaviour is local. If  $\omega_{\text{ns}} = \omega_c$ , there is no no-signalling advantage and this ratio is undefined, so it is not used.

## 4.5 Saturation and quantum achievability

The floor provides a reference against which to compare the behaviours found by the search. For CHSH, where  $\omega_c = 0.75$  and  $\omega_{ns} = 1$ , a score of  $s = 0.80$  requires contextual fraction at least

$$F_{\text{CHSH}}(0.80) = \frac{0.80 - 0.75}{1 - 0.75} = \frac{0.05}{0.25} = 0.20.$$

A behaviour reaching that score may have more contextuality. The floor expresses a necessary amount; whether quantum strategies attain it is a separate question.

For the standard optimal CHSH behaviour, the known score is  $s = (2 + \sqrt{2})/4$ . Substituting gives

$$F_{\text{CHSH}}(s) = \sqrt{2} - 1 \simeq 0.414214,$$

which equals that behaviour's contextual fraction. Here the floor is attained, providing a benchmark for the comparisons across the catalogue.

More generally, equality in the strengthened bound has a direct interpretation in the local decomposition. When both components have positive weight, the local part must score  $\omega_c$  and the no-signalling remainder must score  $\omega_{ns}$ . Both lower limits used in the derivation are then attained.

There is no cancellation that can compensate for an underperforming component, because neither component can beat its own allowed maximum. If one component has zero weight, only the other component's score matters. For a score above  $\omega_c$ , equality in this performance bound is equivalent to the behaviour lying on its contextuality floor.

**A possible direction for discussion.** For which games can a quantum strategy reach the best quantum score using exactly the contextuality required by the floor? Does there exist a quantum-optimal behaviour  $e^*$  with  $\text{CF}(e^*) = F_G(\omega_q)$ , when  $\omega_{ns} > \omega_c$ ?

The remainder  $e_R$  need not itself be quantum-realizable, so the decomposition alone does not guarantee that a quantum behaviour can attain the floor at a chosen score. The saved behaviours provide concrete examples through which to explore this issue.

The next section compares their contextual fractions with the floor. The comparison identifies cases of numerical saturation and cases of positive excess, while leaving open whether another quantum behaviour could achieve the same score with less contextuality.

## 5 Evidence about minimum contextuality

### Numerical observations; equality and minimality remain open

The floor analysis compares the located strategy’s contextual fraction with  $F_G(q_{\text{found}})$ . This uses an achieved score, so it does not require identifying that score with the exact quantum value. Both the score and contextual fraction are nevertheless floating-point evaluations.

All located XOR strategies lie within  $10^{-4}$  of this floor. This includes classically satisfiable cases and cases with no detected quantum advantage; those cases should not be counted as independent evidence of a non-trivial quantum equality principle. The strongest examples in the current data are CHSH and the advantaged three-party classes X3 and X5.

For the general two-party games, classes 5166, 7614 and CHSH also lie on the floor numerically. Six classes have positive excess. A subsequent experiment used 15 independent starts for each of those six classes, followed by a penalised search for lower contextual fraction while maintaining the score.

| Class | Floor at target score | Lowest retained CF | Excess   |
|-------|-----------------------|--------------------|----------|
| 5158  | 0.356789              | 0.379787           | 0.022998 |
| 5734  | 0.257742              | 0.310839           | 0.053097 |
| 5735  | 0.292469              | 0.340962           | 0.048493 |
| 5817  | 0.309401              | 0.347151           | 0.037750 |
| 5821  | 0.356789              | 0.379787           | 0.022998 |
| 7606  | 0.356789              | 0.379787           | 0.022998 |

Table 2. Saved multiple-start experiment. The retained score must be within  $10^{-6}$  below the target. The excess column uses the floor at that target, as in the saved CSV; it is not an exact optimal-set minimum. For class 5735 the retained refinement loses about  $2.14 \times 10^{-7}$  in score, within that tolerance.

Every class retained positive excess. For five of the six classes, the penalised refinement fell outside the score tolerance and was rejected; the reported value therefore comes from the first-stage search. Class 5735 accepted a small reduction in contextual fraction, with a small score loss.

**Interpretation.** Repeated searches have not found a strategy on the floor for these six games. This makes them candidates for a genuine restriction on quantum achievability. It does not prove that a different strategy, a mixture of strategies, or a more complete optimisation could not close the gap.

The current search considers a pure shared state chosen as the top eigenvector at each set of projective qubit measurements. It does not explicitly optimise convex mixtures of equally scoring strategies or all states in degenerate optimal eigenspaces. Since contextual fraction is a convex function, mixtures of optimal behaviours may lower it while preserving the score. This is a concrete issue to test before treating the observed excess as intrinsic to the game.

Across the original 16 analysis rows, ten have strengthened-bound failure residual below  $10^{-5}$  in magnitude; the other six have residuals between approximately 0.00287 and 0.00664. Removing the duplicated CHSH entry gives nine near-tight classes out of 15 distinct classes.

## 6 Noise: contextuality and game advantage

### Strategy-dependent numerical observations

The noise model mixes the located behaviour with uniform white noise:

$$e_\eta = (1 - \eta)e + \eta u, \quad 0 \leq \eta \leq 1. \quad (9)$$

Here  $\eta$  is the proportion of global white noise, not a noise rate independently applied to each party. Since  $u$  is local, increasing this mixing cannot create contextuality. The success probability changes linearly with  $\eta$ .

For a strategy initially beating the classical value, the advantage ends at

$$\eta_{\text{adv}} = \frac{p_{\text{win}}(e) - \omega_c}{p_{\text{win}}(e) - p_{\text{win}}(u)}. \quad (10)$$

The contextuality threshold  $\eta_{\text{CF}}$  is estimated by bisection using the contextual-fraction LP. These thresholds describe the selected behaviour; they need not be identical for different equally scoring strategies.

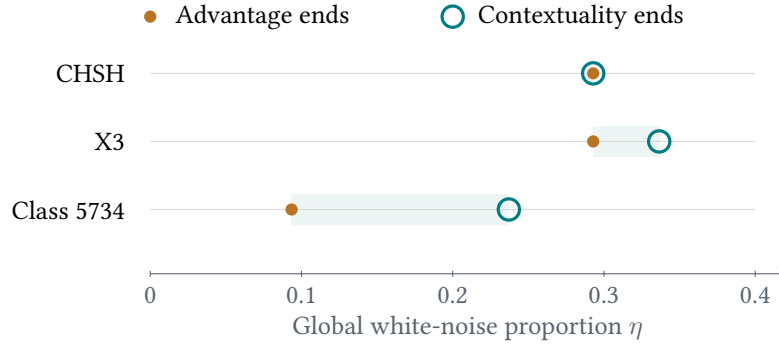


Figure 1. Thresholds from the saved Phase 4 table. Shaded intervals retain contextuality after the game advantage has ended. The two CHSH markers coincide to the precision shown.

For CHSH the benchmark thresholds agree at  $1 - 1/\sqrt{2} \simeq 0.292893$ . For the saved X3 strategy,  $\eta_{\text{adv}} \simeq 0.292893$  while  $\eta_{\text{CF}} \simeq 0.336545$ . Class 5734 has a wider separation: approximately 0.093461 and 0.237130. Values are rounded observations; the bisection interval tolerance is  $10^{-6}$ .

This shows why contextuality and usefulness for a specified game should be kept distinct. A contextual noisy behaviour may fail to beat that game's classical value while violating another Bell inequality. In particular, saturation at the original optimum does not imply saturation along its whole white-noise path: X3 illustrates this in the saved data.

### 6.1 Constructing each noisy table

The noise parameter  $\eta$  can be any number from zero to one:  $\eta = 0$  leaves the original behaviour unchanged, while  $\eta = 1$  leaves only uniform noise. For two parties with binary outputs, every cell of  $u$  is  $1/4$ . The mixture is formed cell by cell, and its score averages in the same way:

$$\begin{aligned} p_\eta(ab \mid xy) &= (1 - \eta)p_e(ab \mid xy) + \eta/4, \\ p_{\text{win}}(e_\eta) &= (1 - \eta)p_{\text{win}}(e) + \eta p_{\text{win}}(u). \end{aligned}$$

Thus  $\eta = 0.2$  means 80% original behaviour and 20% uniform noise. Every sampled point starts from the same original table  $e$ ; noise does not accumulate between points.

## 6.2 The winning probability of uniform noise

Uniform noise gives equal probabilities to the outputs. Its chance of winning depends on which outputs the game accepts in each context. For an illustrative two-party game:

| Inputs $xy$ | Winning outputs $ab$ | Winning probability under $u$ |
|-------------|----------------------|-------------------------------|
| 00          | 00                   | 1/4                           |
| 01          | 00, 11               | 2/4                           |
| 10          | 00, 01, 10           | 3/4                           |
| 11          | 11                   | 1/4                           |

Each cell still has probability  $1/4$ . With equally likely input pairs, the overall winning probability is the average of the four row scores:

$$p_{\text{win}}(u) = \frac{1}{4} \left( \frac{1}{4} + \frac{2}{4} + \frac{3}{4} + \frac{1}{4} \right) = \frac{7}{16} = 43.75\%.$$

Equivalently, seven of the sixteen equally likely input–output combinations win. CHSH accepts two outputs in every row, so its uniform noise score is  $p_{\text{win}}(u) = 1/2$ .

## 6.3 CHSH: scores, contextual fractions and thresholds

The standard optimal quantum CHSH strategy wins with probability  $(2 + \sqrt{2})/4 \simeq 0.853553$ , above the classical bound  $3/4$ . At  $\eta = 0.2$ , for example,

$$p_{\text{win}}(e_{0.2}) = 0.8(0.853553) + 0.2(0.5) \simeq 0.782843,$$

so the noisy strategy still beats the classical score. Along this particular CHSH mixture,

$$p_{\text{win}}(e_\eta) = \frac{1}{2} + \frac{1-\eta}{2\sqrt{2}}, \quad \text{CF}(e_\eta) = \max\{0, \sqrt{2}(1-\eta) - 1\}.$$

| Noise $\eta$                     | Winning probability | CF( $e_\eta$ ) | Beats classical? |
|----------------------------------|---------------------|----------------|------------------|
| 0                                | 85.36%              | 0.4142         | Yes              |
| 0.1                              | 81.82%              | 0.2728         | Yes              |
| 0.2                              | 78.28%              | 0.1314         | Yes              |
| $1 - 1/\sqrt{2} \simeq 0.292893$ | 75.00%              | 0              | Ties             |
| 0.3                              | 74.75%              | 0              | No               |
| 0.5                              | 67.68%              | 0              | No               |
| 1                                | 50.00%              | 0              | No               |

Both thresholds are therefore  $1 - 1/\sqrt{2}$ , about 29.3% noise, for this strategy. In general, `noise_sweep` recalculates CF by LP at each point; it does not assume the CHSH formula or coincident thresholds.

**Sampling and threshold calculation.** The caller chooses the noise grid. The plotting script uses  $0, 0.02, \dots, 0.60$ ; the regression test uses  $0, 0.1, \dots, 0.5$ . A grid such as  $0, 0.1, \dots, 1$  is equally valid. The thresholds are calculated separately, so their precision is not limited by these steps.

`advantage_threshold` solves the linear equation  $p_{\text{win}}(e_\eta) = \omega_c$ . CF cannot increase as more uniform noise is mixed in. For an initially contextual table, `cf_threshold` starts with  $[0, 1]$  and tests its midpoint using the CF LP. If CF is still positive, the threshold lies at a higher noise level; otherwise it lies at or below the midpoint. Repeating this interval-halving procedure is *bisection*. The code treats CF values at most  $10^{-9}$  as zero and stops when the noise interval is no wider than  $10^{-6}$ , giving a numerical estimate.

## 7 Summary and discussion

As of September 2026, the pipeline brings together game enumeration, classical and no-signalling calculations, quantum strategy searches, contextual fraction and noise analysis. The main findings so far are:

- **Game values.** The 65,536 two-party binary winning tables reduce to 805 classes under the relabellings used. Nine classes have a no-signalling advantage over the classical value, and quantum searches find an advantage for all nine.
- **Contextuality and score.** The local decomposition gives a bound on the contextual fraction required for a given score, using the classical and no-signalling values. Several of the strategies found meet this bound numerically. For six two-party classes, the strategies found retain more contextuality than the bound requires. The searches do not establish whether this gap is unavoidable.
- **Noise.** In some examples, contextuality remains after the strategy has lost its advantage over classical strategies in the game being played. This shows that having contextuality and gaining an advantage in a particular game are distinct properties.

These findings combine mathematical deductions with numerical observations, as distinguished in the preceding sections. The points below invite discussion of how to interpret these results, how they relate to existing work and what they suggest for the project.

### Points for discussion

1. Which findings seem most significant, and which existing results would help put them in context?
2. How should the observed differences between contextuality and game performance be understood?
3. Which aspects of the current findings would be most useful to explore further?

## A Supporting data and reproducibility

The repository is [97wfinney/contextual-games](https://github.com/97wfinney/contextual-games). Numbers in Tables 1–2 and Figure 1 can be traced to:

| File under data/                          | Role in this report                                                        |
|-------------------------------------------|----------------------------------------------------------------------------|
| catalogue_2party_general.csv              | 805 classes, orbit sizes and no-signalling screening.                      |
| catalogue_xor2.csv,<br>catalogue_xor3.csv | The XOR class lists and finite classical values.                           |
| catalogue_quantum.csv                     | Explicit-strategy and SDP numerical estimates.                             |
| analysis_phase4.csv                       | Located-strategy scores, contextual fractions, slack and noise thresholds. |
| floor_analysis.csv                        | Contextuality floor at the recorded score and excess.                      |
| multiseed_cf.csv                          | Multiple-start searches and accepted or rejected refinements.              |

**Three-party labels.** The labels X1–X5 used in this report map to canonical IDs as follows; they are shorthand for readability, not an alternative classification convention.

| Label | Canonical ID        |
|-------|---------------------|
| X1    | 7595718147998050665 |
| X2    | 7595718147998050710 |
| X3    | 7595718147998062230 |
| X4    | 7595718148001011305 |
| X5    | 7595718341274539670 |

**Reproduction.** After installing the project dependencies, run `.venv/bin/python -m pytest tests/ -q` for the regression suite. The production sequence is `build_catalogue.py`, `compute_quantum_values.py`, `run_phase4.py`, `floor_analysis.py` and `multiseed_cf.py`, all under `scripts/`. These scripts overwrite their output files. Repeated searches can locate different equally scoring strategies, particularly in the multipartite cases, so noise thresholds need not reproduce identically.

## References

- [1] S. Abramsky and A. Brandenburger, *The sheaf-theoretic structure of non-locality and contextuality*, New Journal of Physics **13**, 113036 (2011). [arXiv:1102.0264](https://arxiv.org/abs/1102.0264).
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- [3] M. Navascués, S. Pironio and A. Acín, *A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations*, New Journal of Physics **10**, 073013 (2008). [arXiv:0803.4290](https://arxiv.org/abs/0803.4290).